

Arav Shah

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Education

Georgia Institute of Technology, BS in Computer Science

Expected May 2027

- Concentrations in AI and HCI. Classes: DSA, Algorithms & Analysis, Knowledge-Based AI, ML, Combinatorics.

Experience

Software Engineering Intern, Société Générale

June 2026 – Aug 2026

- Extended a Java/Python API and Angular/TypeScript trading blotter to support a new order class, integrating order data with Delta exposure across the application stack to support 100+ traders in their daily workflows.
- Delivered end-to-end functionality across Java/TypeScript services, REST APIs, DTO mappings, MongoDB, and PostgreSQL for a stock-index pricing platform processing millions of dollars in daily orders.
- Built comprehensive unit and integration tests covering API contracts, domain-object mappings, database writes, and UI retrieval, reaching 100% test coverage and catching 10+ bugs before deployment.

Software Engineering Intern, Datamaton Inc.

May 2025 – Aug 2025

- Built a JSON REST API and DTO abstraction layer to decouple the C# backend from the UI and enable remote-client integration; deployed with Docker/Kubernetes and an NGINX reverse proxy.
- Refactored a 70K LOC Windows-only C++/CLI .NET Framework business logic library into a cross-platform .NET architecture, removing 10+ OS-dependent components and enabling cross-platform deployment.
- Evaluated multiple YOLO and Darknet-based AI detection models against an existing CNN pipeline, measuring precision, recall, and inference latency on 5,000+ labeled images to select a model for face localization.

Lead Researcher, Georgia Tech AI & Sports Performance Lab

Jan 2025 – May 2026

- Developed models to detect and track players in video footage, measuring player positioning and movement.
- Built a PyTorch experimentation pipeline to preprocess 1,200+ frames, train YOLO models, and compare experiments using precision, recall, mAP, and inference latency to guide model selection and development.
- Increased precision, F1, and mAP by 26% by augmenting data by class and correcting annotation errors.

Projects

TensorQueue - AI Inference Gateway & Job Queue Manager

- Architected a distributed Python/Kubernetes job-processing system for asynchronous AI inference tasks, decoupling a FastAPI client-facing gateway from a scalable worker fleet to support image detection workloads.
- Implemented a persistent task queue utilizing Redis with exponential backoff, a dead-letter queue, and crash recovery via visibility timeouts, enabling fault-tolerant processing of 1.5K+ image-processing tasks/day.
- Engineered a Python/PyTorch worker service to execute YOLO-based object detection, efficiently utilizing hardware and maintaining sub-100ms p99 processing latency via model warm-up and queue-based autoscaling.

Technical Skills

Languages: C++, C#, Java, Python, JavaScript, TypeScript, CSS, SQL, Bash.

Technologies: .NET Core/Framework, Spring Boot, Kubernetes, Docker, NGINX, YOLO, Angular, MongoDB.

Involvement

Infrastructure Lead & Vice President of Marketing, AI@GT – Atlanta, GA

Jan 2024 – Present

- Built a 5-node, 40GB VRAM GPU cluster for shared model training and inference, expanding AI compute infrastructure for student researchers and enabling scalable model training and inference workloads.
- Automated GPU assignment and node monitoring via Python and Bash for continuous cluster status visibility.
- Led marketing and sponsor outreach to expand on-campus AI education, student engagement, & career options.